

A Quantum Resource Theory of Transpose-Reciprocity Breaking from Non-Hermitian Propagators:

Exact Hatano–Nelson Calibration and Uniform Short-Time Persistence in the Single-Particle Thermodynamic Limit

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We formulate a fixed-time quantum resource theory of transpose-reciprocity breaking for channels constructed from normalized non-Hermitian propagators by a canonical Julia unitary completion. The primitive objects are CPTP bistochastic channels on the doubled physical–environmental Hilbert space; free channels are invariant under channel transpose, and free superchannels are physically realizable, object-preserving superchannels that commute with this transpose involution. This covariance is an axiomatic resource-neutrality criterion, not a microscopic hardware cost model. The channel asymmetry $A_\diamond = \frac{1}{2}\|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_\diamond$ is faithful and equals $2p_{\text{succ}}^{\text{opt}} - 1$ in one-shot discrimination of a channel from its reciprocal transpose; the generalized robustness is another faithful monotone.

For the two-site Hatano–Nelson gauge orbit at a calibration time, $A_\diamond$ has a closed form and is strictly increasing in $|g|$ under a common strict normalization. Hence any deterministic free conversion on that orbit must satisfy $|g| \geq |g'|$. An explicitly defined ancilla-free unitary-congruence subclass is rigid, $|g| = |g'|$, while deterministic attenuation for the full ancilla-assisted free-superchannel class is left open.

The quantitative resource value is normalization dependent, whereas its zero-versus-positive support is invariant under every finite strict scalar normalization for the canonical Julia representative. A per-Hamiltonian tight gauge gives exact invariance under overall positive energy rescaling after optimization over all positive times. For uniformly bounded finite-range chains, an L -uniform short-time estimate yields a lower bound in the single-particle thermodynamic limit; for the uniform Hatano–Nelson family this bound is strictly positive for every $g \neq 0$.

The resource detects transpose-reciprocity breaking, not non-Hermiticity or skin accumulation in full generality. An exact biconditional with conventional OBC NHSE holds on model classes where conventional NHSE is independently equivalent to transpose-reciprocity breaking, in particular for the uniform Hatano–Nelson family. For coupled chains we prove exponential buffer stability only for the compressed-contraction Julia channel Λ_V ; no identification with a genuine many-body reduced channel is claimed. Numerical checks support the stated locality and finite-size persistence claims but are not used as proofs.

I. INTRODUCTION AND OPERATIONAL SCOPE

Quantum resource theories specify an object class, a free subset, admissible free transformations, and a conversion preorder; channel resource theories naturally use quantum superchannels for the transformations [1–3]. Throughout, $\|\cdot\|$ denotes the Hilbert-space operator norm $\|\cdot\|_{2 \rightarrow 2}$. Here the primitive resource is not “non-Hermiticity” or NHSE itself, but violation of transpose reciprocity in a fixed physical basis.

The construction is

$$H \longrightarrow U_H(t) \longrightarrow K_{H,t} \longrightarrow \tilde{U}_{K_{H,t}} \longrightarrow \mathcal{N}_{H,t}. \quad (1)$$

where the propagator is divided by a positive scalar so that $K_{H,t}$ is a strict contraction, and the contraction is completed to a canonical Julia unitary. This is a representation convention for the present QRT, not a uniqueness statement about all physical open-system realizations of a non-Hermitian evolution.

The reciprocity datum is a fixed basis. Let $\mathcal{C}(X) = X^*$ be complex conjugation in that basis. For a bistochastic

channel $\mathcal{N}$ with Kraus operators $\{K_a\}$, define

$$\mathcal{N}^{\text{rev}}(X) = \sum_a K_a^T X K_a^*. \quad (2)$$

Then

$$\mathcal{N}^{\text{rev}} = \mathcal{C} \circ \mathcal{N}^\dagger \circ \mathcal{C}, \quad (3)$$

so $\mathcal{N} = \mathcal{N}^{\text{rev}}$ is a time-reversal-like covariance condition in the chosen reciprocity frame. It is related in spirit to, but is not identified with, standard quantum detailed-balance frameworks [14–16]; no equivalence between the two notions is assumed.

A central scope distinction will be maintained throughout:

QRT resource = transpose-reciprocity breaking,

(4)

while NHSE is a distinguished calibration sector. We also keep five mathematically distinct objects separate: the full Julia-channel monotone $A_\diamond$; the physical-input witness $A_\diamond^{\text{phys}}$, for which no full- $\mathfrak{D}_L$ monotonicity is claimed; the compressed single-particle channel $\Lambda_V = \mathcal{N}_{P_V K P_V}$; and, only when an independent tensor-product realization is supplied, a genuine reduced channel $\mathfrak{R}_V$. The

covariance-matrix quantity $\Xi_V := d_V^{-1} K_{V\bar{V}} K_{V\bar{V}}^\dagger$ is a separate diagnostic, not a CPTP reduced-channel component. No identification among these objects is assumed beyond the relations proved explicitly below. The resource can be positive for Hermitian, gauge-trivial systems that violate transpose reciprocity in the fixed frame; conversely, other NHSE mechanisms, including reciprocal non-Abelian $\mathbb{Z}_2$ skin effects and genuinely Lindbladian skin topology, lie outside this single-involution construction [9, 19, 20].

II. CANONICAL JULIA CHANNEL AND TRANSPOSE COVARIANCE

Definition 1 (Single-particle Hilbert space). *For a chain of length L with internal dimension M ,*

$$\mathcal{H}_L = \text{span}\{|n, \alpha\rangle : 1 \leq n \leq L, 1 \leq \alpha \leq M\}, \quad (5)$$

with a fixed spatial and internal basis. The canonical Julia completion acts on $\tilde{\mathcal{H}}_L = \mathcal{H}_L \oplus \mathcal{H}_L^{\text{env}}$, where the environmental copy carries the corresponding basis.

Definition 2 (Admissible normalized propagator). *Let $\mathcal{H}_{\text{adm}}$ be a transpose-closed Hamiltonian class with a uniform operator-norm bound $\sup_{H \in \mathcal{H}_{\text{adm}}} \|H\| \leq h_{\text{adm}} < \infty$, and let $U_H(t) = e^{-iHt}$. A common scalar normalization is*

$$\kappa_\varepsilon(t) = [1 + \varepsilon(t)] \kappa_*(t), \quad \kappa_*(t) = \sup_{H \in \mathcal{H}_{\text{adm}}} \|U_H(t)\|^2. \quad (6)$$

Then $\kappa_(t) \leq e^{2h_{\text{adm}}t} < \infty$, so the common normalization is well defined. Here $\varepsilon(t) > 0$.*

$$K_{H,t} = \frac{U_H(t)}{\sqrt{\kappa_\varepsilon(t)}}, \quad \|K_{H,t}\| < 1, \quad (7)$$

and, because transpose preserves the operator norm,

$$K_{H^T,t} = K_{H,t}^T. \quad (8)$$

Proposition 1 (Julia completion). *For every strict contraction K , define*

$$D_K = (\mathbb{1} - K^\dagger K)^{1/2}, \quad D_{K^\dagger} = (\mathbb{1} - K K^\dagger)^{1/2}, \quad (9)$$

and

$$\tilde{U}_K = \begin{pmatrix} K & D_{K^\dagger} \\ D_K & -K^\dagger \end{pmatrix}. \quad (10)$$

Then $\tilde{U}_K$ is unitary. Moreover,

$$D_K^T = D_{K^*}, \quad D_{K^\dagger}^T = D_{K^T}, \quad \tilde{U}_K^T = \tilde{U}_{K^T}. \quad (11)$$

Consequently the canonical Julia channel

$$\mathcal{N}_K(X) = \tilde{U}_K X \tilde{U}_K^\dagger \quad (12)$$

is CPTP, unital, and obeys

$$\mathcal{N}_K^{\text{rev}} = \mathcal{N}_{K^T}. \quad (13)$$

Proof. Unitarity follows from the standard defect identities $D_{K^\dagger} K = K D_K$ and $D_K K^\dagger = K D_{K^\dagger}$. The transpose identities follow by functional calculus. The channel statement follows from the transposed implementing unitary. $\square$

Corollary 1 (Canonical reciprocity criterion). *For a strict contraction K ,*

$$\mathcal{N}_K^{\text{rev}} = \mathcal{N}_K \iff K^T = K. \quad (14)$$

Proof. The forward implication follows because equality of unitary-conjugation channels implies $\tilde{U}_K^T = e^{i\phi} \tilde{U}_K$. Transposition gives $e^{2i\phi} = 1$. The choice $e^{i\phi} = -1$ would require a positive definite defect operator to equal the negative of another positive definite defect operator, which is impossible for a strict contraction. Hence $e^{i\phi} = 1$ and $K^T = K$. $\square$

Remark 1 (What is dilation independent). *The canonical criterion (14) is a statement about the chosen Julia representative. For an arbitrary unitary dilation V with physical block $V_{11} = K$, one can only conclude*

$$V^T \propto V \implies K^T = \pm K \quad (15)$$

(for $K \neq 0$). Thus the correct universal dilation-independence statement is

$$\begin{aligned} & K^T \neq \pm K \\ \implies & \text{every unitary dilation of } K \text{ is non-reciprocal.} \end{aligned} \quad (16)$$

The stronger implication with $K^T \neq K$ is false in general; the canonical Julia construction is therefore kept explicit throughout the paper.

Definition 3 (Fixed-time resource channel). *The primitive object associated with (H, t) is*

$$\mathcal{N}_{H,t} = \mathcal{N}_{K_{H,t}}. \quad (17)$$

The resource depends on the chosen scalar normalization and reciprocity frame.

III. FREE CHANNELS, FREE SUPERCHANNELS, AND MONOTONES

Definition 4 (Finite-size object class and free set). *Let*

$$\text{Obj}_L = \{\mathcal{N} \in \text{Chan}(\tilde{\mathcal{H}}_L) : \mathcal{N}(\mathbb{1}) = \mathbb{1}\}, \quad (18)$$

namely the CPTP bistochastic channels on the doubled space, and define

$$\mathfrak{F}_L = \{\mathcal{N} \in \text{Obj}_L : \mathcal{N}^{\text{rev}} = \mathcal{N}\}. \quad (19)$$

At fixed finite L this is a symmetry resource theory. Thermodynamic locality is imposed only through L -uniform family conditions.

Lemma 1 (Basic properties of channel transpose). *On Obj_L , the map $\mathcal{N} \mapsto \mathcal{N}^{\text{rev}}$ is a linear continuous involution, preserves CPTP bistochasticity, and satisfies*

$$(\mathcal{A} \circ \mathcal{B})^{\text{rev}} = \mathcal{B}^{\text{rev}} \circ \mathcal{A}^{\text{rev}}, \quad (\mathcal{N} \otimes \mathbb{1}_E)^{\text{rev}} = \mathcal{N}^{\text{rev}} \otimes \mathbb{1}_E. \quad (20)$$

Proof. For Kraus operators K_a , use $(K_a^T)^\dagger = K_a^*$ and bistochasticity to verify trace preservation and unitality. Kraus-representation independence follows from the usual unitary Kraus mixing. The anti-homomorphism property follows from $(AB)^T = B^T A^T$. $\square$

Definition 5 (Free superchannel). *A deterministic superchannel has the form*

$$\Theta(\mathcal{N}) = \Gamma_{\text{post}} \circ (\mathcal{N} \otimes \mathbb{1}_E) \circ \Gamma_{\text{pre}}, \quad (21)$$

with CPTP pre- and post-processing [3]. It is free when

1. Θ is physically realizable,
2. $\Theta(\text{Obj}_L) \subseteq \text{Obj}_L$,
3. $\Theta(\mathcal{N}^{\text{rev}}) = [\Theta(\mathcal{N})]^{\text{rev}}$ for all $\mathcal{N} \in \text{Obj}_L$.

The covariance requirement is imposed on the complete superchannel. We denote the class by $\mathfrak{D}_L$.

Remark 2 (Physical status of the free-operation choice). *The class $\mathfrak{D}_L$ is an axiomatic covariance class. It guarantees resource non-generation, but it is not claimed that every mathematically free superchannel is zero-cost in a specific laboratory technology. A hardware-level resource theory would require an independently specified set of symmetry-neutral fabrication and control primitives. The present results therefore establish resource-theoretic monotonicity for the chosen symmetry class, rather than a microscopic zero-cost theorem for laboratory engineering.*

Definition 6 (Channel asymmetry).

$$A_\diamond(\mathcal{N}) = \frac{1}{2} \|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_\diamond. \quad (22)$$

Proposition 2 (Operational discrimination). *For equal priors, the optimal one-shot success probability for discriminating $\mathcal{N}$ from $\mathcal{N}^{\text{rev}}$ is*

$$p_{\text{succ}}^{\text{opt}} = \frac{1}{2} + \frac{1}{4} \|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_\diamond = \frac{1 + A_\diamond(\mathcal{N})}{2}. \quad (23)$$

Hence

$$A_\diamond = 2p_{\text{succ}}^{\text{opt}} - 1. \quad (24)$$

Theorem 1 (Monotonicity and faithfulness). *For every $\Theta \in \mathfrak{D}_L$,*

$$A_\diamond(\Theta(\mathcal{N})) \leq A_\diamond(\mathcal{N}), \quad (25)$$

and

$$A_\diamond(\mathcal{N}) = 0 \iff \mathcal{N} \in \mathfrak{F}_L. \quad (26)$$

Proof. Covariance yields $[\Theta(\mathcal{N})]^{\text{rev}} = \Theta(\mathcal{N}^{\text{rev}})$. Contractivity of the diamond norm under physical superchannels then gives the inequality; faithfulness is immediate. $\square$

Definition 7 (Generalized robustness).

$$R_{\text{rob}}(\mathcal{N}) = \inf \left\{ s \geq 0 : \exists \mathcal{M} \in \text{Obj}_L, \frac{\mathcal{N} + s\mathcal{M}}{1+s} \in \mathfrak{F}_L \right\}. \quad (27)$$

Theorem 2 (Robustness monotonicity and boundedness). *For every free superchannel,*

$$\begin{aligned} R_{\text{rob}}(\Theta(\mathcal{N})) &\leq R_{\text{rob}}(\mathcal{N}), \\ 0 &\leq A_\diamond(\mathcal{N}) \leq R_{\text{rob}}(\mathcal{N}) \leq 1. \end{aligned} \quad (28)$$

Moreover,

$$R_{\text{max}}(\mathcal{N}) = \log[1 + R_{\text{rob}}(\mathcal{N})] \quad (29)$$

with the max-relative-entropy form

$$R_{\text{max}}(\mathcal{N}) = \min_{\mathcal{F} \in \mathfrak{F}_L} \log \min \{ \lambda \geq 1 : \lambda \mathcal{F} - \mathcal{N} \text{ is CP} \}. \quad (30)$$

Consequently $R_{\text{rob}}(\mathcal{N}^{\otimes n}) \leq 1$ and $R_{\text{max}}(\mathcal{N}^{\otimes n}) \leq \log 2$ for all n when the free set on the tensor-power space is defined by the same involution.

Proof. Free mixtures are mapped to free mixtures by a free superchannel, giving monotonicity. The bound $R_{\text{rob}} \leq 1$ follows by choosing $\mathcal{M} = \mathcal{N}^{\text{rev}}$, since $(\mathcal{N} + \mathcal{N}^{\text{rev}})/2 \in \mathfrak{F}_L$; the same identity gives $A_\diamond \leq R_{\text{rob}}$. The R_{max} equivalence follows by setting $s = \lambda - 1$ and observing that a CP, TP, unital map divided by s belongs to Obj_L . The tensor-power bound follows from $\frac{1}{2}[\mathcal{N}^{\otimes n} + (\mathcal{N}^{\text{rev}})^{\otimes n}] \in \mathfrak{F}_L$. $\square$

Definition 8 (Physical-input asymmetry). *The full $A_\diamond$ is the QRT monotone of the complete canonical Julia channel. To describe laboratory probes that cannot prepare the environmental input block, let P_{phys} denote the projector onto $\mathcal{H}_L$ and let the reference system satisfy $\dim R \geq \dim \mathcal{H}_L$. Define*

$$A_\diamond^{\text{phys}}(\mathcal{N}) = \sup_{\rho \in \mathcal{D}(\mathcal{H}_L \otimes R)} \frac{1}{2} \|\Delta_\rho\|_1, \quad (31)$$

with

$$\Delta_\rho = (\mathcal{N} \otimes \mathbb{1}_R - \mathcal{N}^{\text{rev}} \otimes \mathbb{1}_R)(\rho). \quad (32)$$

where the input is embedded into the physical block selected by P_{phys} , with no support on the environmental input block. This restricted quantity is an operational witness; no monotonicity under the full $\mathfrak{D}_L$ class is claimed.

Proposition 3 (Physical-input probe). *For the canonical Julia channel,*

$$A_\diamond^{\text{phys}}(\mathcal{N}_K) \geq \frac{1}{2} \max_m \sum_n ||K_{nm}|^2 - |K_{mn}|^2|. \quad (33)$$

Under strict contraction, $A_\diamond^{\text{phys}} = 0$ iff $K^T = K$, so the physical-input witness has the same support as the canonical full-channel monotone.

Proof. For the lower bound, use the physical basis input $|m\rangle$ and measure the physical site basis. Under $\mathcal{N}_K$ and $\mathcal{N}_K^{\text{rev}}$ the corresponding physical probabilities are $|K_{nm}|^2$ and $|K_{mn}|^2$, respectively, so data processing gives Eq. (33). For the support statement, suppose $A_\diamond^{\text{phys}} = 0$. Then, in particular, for every pure physical input $|\psi\rangle$, the two pure output states of the isometries $\tilde{U}_K P_{\text{phys}}$ and $\tilde{U}_K^T P_{\text{phys}}$ coincide as density operators. If $\dim \mathcal{H}_L > 1$, linearity implies that the two isometries differ on the physical input subspace by one common phase, $\tilde{U}_K^T P_{\text{phys}} = e^{i\phi} \tilde{U}_K P_{\text{phys}}$. For $\dim \mathcal{H}_L = 1$, $K^T = K$ is automatic. Comparing the physical diagonal and defect blocks gives $K^T = e^{i\phi} K$ and $D_K^T = e^{i\phi} D_K$. Strict contraction makes both defect operators positive definite, so $e^{i\phi} = -1$ is impossible; hence $e^{i\phi} = 1$ and $K^T = K$. The converse follows from Eq. (14). Finally, $A_\diamond^{\text{phys}} \leq A_\diamond$ because the former optimizes over a restricted set of inputs. $\square$

IV. NORMALIZATION GAUGE AND TRAJECTORY RESOURCE

Definition 9 (Admissible scalar gauge). *A positive scalar $\kappa(H, t)$ is admissible when*

$$\kappa(H, t) > \|U_H(t)\|^2, \quad \kappa(H^T, t) = \kappa(H, t), \quad (34)$$

and, on thermodynamic families of interest, it is uniformly bounded above and below on compact time intervals. When a short-time expansion is used,

$$\kappa(H_L, t) = \kappa_0 + O(t) \quad (35)$$

uniformly in L .

Proposition 4 (Gauge-independent support). *For every finite-dimensional H, t and every admissible scalar gauge,*

$$A_\diamond(\mathcal{N}_{H,t}^{(\kappa)}) = 0 \iff U_H(t)^T = U_H(t). \quad (36)$$

Thus the zero-versus-positive support is gauge independent within the finite strict scalar class.

Proof. Because the normalization is scalar, $[K_{H,t}^{(\kappa)}]^T = K_{H,t}^{(\kappa)}$ iff $U_H(t)^T = U_H(t)$; then use Eq. (14). $\square$

Proposition 5 (Quantitative gauge continuity). *Suppose two gauges give strict contractions obeying $\|K_j\| \leq q < 1$. Then*

$$\begin{aligned} & |A_\diamond(\mathcal{N}_{K_1}) - A_\diamond(\mathcal{N}_{K_2})| \\ & \leq 2 \left(1 + \frac{q}{\sqrt{1-q^2}} \right) \|K_1 - K_2\|. \end{aligned} \quad (37)$$

For the same propagator $U_H(t)$,

$$\|K_1 - K_2\| = \|U_H(t)\| \left| \kappa_1^{-1/2} - \kappa_2^{-1/2} \right|. \quad (38)$$

Proposition 6 (Large-normalization degeneration). *For fixed H, t ,*

$$K^{(\kappa)} = U_H(t)/\sqrt{\kappa} \longrightarrow 0, \quad \tilde{U}_{K^{(\kappa)}} \longrightarrow \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \quad (39)$$

so

$$A_\diamond(\mathcal{N}_{K^{(\kappa)}}) \longrightarrow 0 \quad (40)$$

as $\kappa \rightarrow \infty$. Quantitative resource values are therefore gauge dependent.

Definition 10 (Trajectory resources and tight gauge). *For an admissible scalar gauge κ , define the trajectory-derived resource*

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) = \sup_{t \in \mathcal{T}} A_\diamond(\mathcal{N}_{H,t}^{(\kappa)}). \quad (41)$$

Fix $\varepsilon_0 > 0$ and define the per-Hamiltonian tight gauge by

$$\kappa_H^{\text{tight}}(t) = (1 + \varepsilon_0) \|U_H(t)\|^2. \quad (42)$$

Then $\|K_{H,t}^{\text{tight}}\| = q_{\text{tight}} = (1 + \varepsilon_0)^{-1/2} < 1$. For $\mathcal{T} = (0, \infty)$ write

$$A_\diamond^{\text{sup}}(H) = A_{\diamond, \text{tight}}^{\text{sup}}(H; (0, \infty)). \quad (43)$$

Proposition 7 (Exact energy-rescaling covariance). *For every $\lambda > 0$,*

$$\mathcal{N}_{\lambda H, t}^{\text{tight}} = \mathcal{N}_{H, \lambda t}^{\text{tight}}, \quad A_\diamond^{\text{sup}}(\lambda H) = A_\diamond^{\text{sup}}(H). \quad (44)$$

Proof. Use $U_{\lambda H}(t) = U_H(\lambda t)$ and the definition of the tight normalization. $\square$

Theorem 3 (Trajectory characterization of transpose-reciprocity breaking). *Let the operational time set contain a sequence $t_j \downarrow 0$. For every admissible strict scalar gauge κ ,*

$$A_{\diamond, \kappa}^{\text{sup}}(H; \mathcal{T}) > 0 \iff H^T \neq H. \quad (45)$$

In particular, the statement holds for the tight gauge.

Proof. If $H^T = H$, then $U_H(t)^T = U_H(t)$ for all t . Conversely, $U_H(t)^T - U_H(t) = -i(H^T - H)t + O(t^2)$, so for sufficiently small positive t the two propagators differ. Gauge-independent support then gives a strictly positive channel asymmetry. $\square$

V. EXACT HATANO-NELSON CALIBRATION AND CONVERSION CONSTRAINTS

Consider

$$H_g = m \mathbb{1} + t_0(e^g |1\rangle\langle 2| + e^{-g} |2\rangle\langle 1|), \quad m, t_0 \in \mathbb{R}, \quad t_0 > 0, \quad (46)$$

with calibration time

$$\tau_* = \frac{\pi}{2t_0}. \quad (47)$$

Up to a global phase,

$$U_g(\tau_*) = -i \begin{pmatrix} 0 & e^g \\ e^{-g} & 0 \end{pmatrix}. \quad (48)$$

With a common normalization $\kappa > e^{2g_{\max}}$ on $|g| \leq g_{\max}$, put $u = \kappa^{-1/2}$. Then

$$K_g = -iu \begin{pmatrix} 0 & e^g \\ e^{-g} & 0 \end{pmatrix}, \quad D_{K_g} = \text{diag}(a, b), \quad (49)$$

where

$$a = \sqrt{1 - u^2 e^{-2g}}, \quad b = \sqrt{1 - u^2 e^{2g}}. \quad (50)$$

Theorem 4 (Exact two-site channel asymmetry). *At τ_* ,*

$$\boxed{A_\diamond(\mathcal{N}_{g,\tau_*}) = \sqrt{1 - c(g)^2}, \quad c(g) = u^2 + \sqrt{(1 - u^2 e^{2g})(1 - u^2 e^{-2g})}.} \quad (51)$$

For $g > 0$,

$$\boxed{\frac{d}{dg} A_\diamond(\mathcal{N}_{g,\tau_*}) > 0.} \quad (52)$$

Proof. The discrimination relative unitary $W_g = \tilde{U}_{K_g}^\dagger \tilde{U}_{K_g}^T$ has eigenvalues $e^{\pm i\theta(g)}$, each twice, with

$$\cos \theta(g) = u^2 + ab = c(g). \quad (53)$$

The exact unitary-channel discrimination formula gives $A_\diamond = \sqrt{1 - \nu(W_g)^2}$, where ν is the distance from the origin to the numerical range of W_g [11]. In the strict calibration regime the numerical range is the segment joining $e^{\pm i\theta}$, hence $\nu = c(g)$. Writing $y = e^{2g}$ gives

$$c'(g) = -\frac{u^2(y - y^{-1})}{\sqrt{1 + u^4 - u^2(y + y^{-1})}} < 0, \quad (54)$$

so $A_\diamond$ is strictly increasing. $\square$

Corollary 2 (Necessary conversion order). *For the common normalization,*

$$\boxed{|g| \geq |g'| \iff A_\diamond(\mathcal{N}_{g,\tau_*}) \geq A_\diamond(\mathcal{N}_{g',\tau_*}).} \quad (55)$$

Therefore

$$\boxed{\mathcal{N}_{g,\tau_*} \succeq_{\mathfrak{D}} \mathcal{N}_{g',\tau_*} \implies |g| \geq |g'|.} \quad (56)$$

The converse is not established.

Definition 11 (Ancilla-free unitary-congruence subclass). *For a unitary V , let $\mathcal{V}(X) = VXV^\dagger$ and define*

$$\Theta_V(\mathcal{N}) = \mathcal{V}^{\text{rev}} \circ \mathcal{N} \circ \mathcal{V}, \quad \mathcal{V}^{\text{rev}}(X) = V^T X V^*. \quad (57)$$

We call this the unitary-congruence subclass and denote by $\mathfrak{D}_{\text{uc}}$ those members that preserve the object class.

Lemma 2 (Cosquare invariant). *For a unitary U define*

$$C_{\text{cs}}(U) = U^* U, \quad \chi_{\text{cs}}(U) = \frac{1}{d^2} |\text{Tr}(U^* U)|^2. \quad (58)$$

Then χ_{cs} is invariant under $U \mapsto e^{i\phi} V^T U V$. For the two-site Julia unitary,

$$\boxed{\chi_{\text{cs}}(\tilde{U}_{K_g}) = c(g)^2.} \quad (59)$$

Theorem 5 (Rigidity in the unitary-congruence subclass). *If $\Theta_V \in \mathfrak{D}_{\text{uc}}$ maps $\mathcal{N}_{g,\tau_*}$ to $\mathcal{N}_{g',\tau_*}$, then*

$$\boxed{|g| = |g'|.} \quad (60)$$

Proof. Output unitary channels differ only by a global phase, so the cosquare invariant is unchanged. Thus $c(g)^2 = c(g')^2$, and positivity plus strict monotonicity in $|g|$ gives the result. $\square$

Remark 3 (Conversion status). *The results distinguish three statements. The monotone proves the necessary attenuation order $|g| \geq |g'|$ for the full free-superchannel class; the unitary-congruence subclass is rigid; and explicit deterministic attenuation to the particular target channel $\mathcal{N}_{g',\tau_*}$ remains an open feasibility problem for the full ancilla-assisted class. No sufficiency claim is made.*

VI. EXACT SCOPE OF THE NHSE CORRESPONDENCE

For a clean nearest-neighbor translation-invariant chain,

$$H(\beta) = H_{-1}\beta^{-1} + H_0 + H_1\beta. \quad (61)$$

Define the GBZ radius by

$$\boxed{\rho_{\text{GBZ}} = \frac{1}{a} \sup_{\beta \in \mathcal{C}_{\text{GBZ}}} |\ln |\beta||.} \quad (62)$$

Here “conventional OBC NHSE” denotes extensive boundary accumulation of OBC eigenstates in the standard non-Bloch sense; for the scalar Hatano–Nelson family this is equivalent to $|g| > 0$. If

$$H_{-1} = H_1^T, \quad H_0 = H_0^T, \quad (63)$$

then

$$H(\beta)^T = H(\beta^{-1}), \quad (64)$$

so zeros of the characteristic equation occur in reciprocal pairs. For a point-gap energy E with $\det[H(k) - E] \neq 0$ for all real k , the determinant winding therefore vanishes,

$$W_{\text{BZ}}(E) = \frac{1}{2\pi i} \int_{-\pi}^{\pi} dk \partial_k \log \det[H(k) - E] = 0. \quad (65)$$

More specifically, assume H_1 and H_{-1} are invertible so that the nearest-neighbor characteristic equation has degree $2M$ with finite nonzero roots counted with multiplicity, and assume the standard generic GBZ root-ordering prescription applies at E . If $|\beta_1| \leq \dots \leq |\beta_{2M}|$ and the GBZ condition is $|\beta_M| = |\beta_{M+1}|$, reciprocal pairing implies $|\beta_j| |\beta_{2M+1-j}| = 1$ and hence $|\beta_M| = |\beta_{M+1}| = 1$, giving $\rho_{\text{GBZ}} = 0$. This is a conditional multichannel statement under the displayed hypotheses, not a universal implication from reciprocity alone.

For the scalar Hatano–Nelson family,

$$H_g(\beta) = m + Je^g\beta + Je^{-g}\beta^{-1}, \quad (66)$$

and the two roots obey $\beta_1\beta_2 = e^{-2g}$, so

$$\boxed{\rho_{\text{GBZ}} = |g|/a.} \quad (67)$$

Under OBC,

$$H_{g,L} = S_g H_{0,L} S_g^{-1}, \quad S_g |n\rangle = e^{-gn} |n\rangle, \quad (68)$$

so the right eigenvectors satisfy

$$\psi_{j,g}(n) \propto e^{-gn} \sin \frac{\pi j n}{L+1}. \quad (69)$$

Thus the uniform Hatano–Nelson family obeys

$$\boxed{g \neq 0 \iff H_{g,L}^T \neq H_{g,L} \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC NHSE.}} \quad (70)$$

Corollary 3 (Exact HN resource witness). *With operational time domain $(0, \infty)$ and the tight gauge,*

$$\boxed{A_\diamond^{\text{sup}}(H_g) > 0 \iff g \neq 0 \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC NHSE.}} \quad (71)$$

This is an exact result for the uniform Hatano–Nelson family, not a universal theorem for all NHSE mechanisms.

Proposition 8 (Hermitian counterexample to a universal NHSE interpretation). *Let $H = H^\dagger$ but $H^T \neq H$. Then, for any time domain accumulating at zero, $A_\diamond^{\text{sup}}(H) > 0$. In particular,*

$$H_\phi = J \sum_{n=1}^{L-1} (e^{i\phi} |n+1\rangle \langle n| + e^{-i\phi} |n\rangle \langle n+1|) \quad (72)$$

has positive trajectory resource for $\phi \notin \pi\mathbb{Z}$, even though $H_\phi = DH_0 D^\dagger$ with diagonal unitary D and therefore has the same site-intensity profiles as the real symmetric chain H_0 .

Remark 4. *The exact logical hierarchy is therefore*

$$H^T \neq H \iff A_\diamond^{\text{sup}} > 0, \quad (73)$$

while an NHSE biconditional requires an additional model-class assumption that conventional NHSE is equivalent to transpose-reciprocity breaking. Reciprocal non-Abelian $\mathbb{Z}_2$ skin physics is outside this class [9, 19].

VII. UNIFORM SHORT-TIME PERSISTENCE IN THE SINGLE-PARTICLE THERMODYNAMIC LIMIT

Assumption 1 (Uniform finite-range family). *A family $\{H_L\}$ satisfies*

$$\sup_L \|H_L\| \leq h < \infty, \quad (74)$$

and has uniformly bounded finite interaction range.

Lemma 3 (Uniform propagator remainder). *Under the assumption,*

$$U_L(t) = \mathbb{1} - iH_L t + R_L(t), \quad \boxed{\|R_L(t)\| \leq \frac{1}{2} h^2 t^2 e^{ht}.} \quad (75)$$

Proof. Use the exponential series and $\sum_{r \geq 2} (ht)^r / r! \leq \frac{1}{2} h^2 t^2 e^{ht}$. $\square$

Definition 12 (Transition-probability asymmetry defect).

$$\boxed{D_{\text{amp}}(H) = \max_m \sum_{n \neq m} ||H_{nm}|^2 - |H_{mn}|^2|.} \quad (76)$$

Remark 5 (Scope of the local witness). *$D_{\text{amp}}(H) > 0$ is a sufficient condition for the quadratic short-time lower bound below, but it is not a characterization of transpose-reciprocity breaking. In particular, phase-only or gauge-type reciprocity breaking may have $|H_{nm}| = |H_{mn}|$ for all n, m and hence $D_{\text{amp}}(H) = 0$ while $H^T \neq H$.*

Lemma 4 (Uniform transition-intensity expansion). *There is an L -independent function*

$$\mathcal{E}_h(t) = h^3 t^3 e^{ht} + \frac{1}{4} h^4 t^4 e^{2ht} \quad (77)$$

such that

$$\max_m \sum_{n \neq m} ||U_L(t)_{nm}|^2 - |U_L(t)_{mn}|^2| - t^2 (|(H_L)_{nm}|^2 - |(H_L)_{mn}|^2) \leq 2\mathcal{E}_h(t). \quad (78)$$

Proof. Write $U_{nm} = -it(H_L)_{nm} + R_{nm}$ and use $|x+y|^2 - |x|^2 \leq 2|x||y| + |y|^2$, followed by Cauchy–Schwarz and the previous lemma. Apply the same argument to $U_L^\dagger$ for the row estimate. $\square$

Proposition 9 (Uniform finite-size short-time resource bound). *Let the scalar normalization satisfy $\kappa_L(t) = \kappa_0 + O(t)$ uniformly in L , with $\kappa_0 > 0$. Then there are $t_0, C_* > 0$, independent of L , such that*

$$\boxed{A_\diamond(\mathcal{N}_{H_L,t}) \geq \frac{t^2}{2\kappa_0} D_{\text{amp}}(H_L) - C_* t^3, \quad 0 < t < t_0.} \quad (79)$$

Proof. Use the physical-input probe Eq. (33). The transition-intensity lemma gives the numerator lower bound. Since $\kappa_L(t)^{-1} = \kappa_0^{-1} + O(t)$ uniformly and $D_{\text{amp}}(H_L)$ is uniformly bounded for a finite-range uniformly bounded family, all corrections are L -independent $O(t^3)$ terms. $\square$

Corollary 4 (Short-time persistence in the single-particle thermodynamic limit). *With*

$$A_{\diamond,\infty}(t) = \limsup_{L \rightarrow \infty} A_{\diamond}(\mathcal{N}_{H_L}, t), \quad (80)$$

one has

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond,\infty}(t)}{t^2} \geq \frac{1}{2\kappa_0} \limsup_{L \rightarrow \infty} D_{\text{amp}}(H_L). \quad (81)$$

Proposition 10 (Uniform Hatano–Nelson persistence). *For*

$$H_{g,L} = m\mathbb{1} + J e^g T_+ + J e^{-g} T_-, \quad T_- = T_+^T, \quad (82)$$

$$\|H_{g,L}\| \leq |m| + 2J \cosh |g| =: h_g, \quad (83)$$

and, for $L \geq 3$,

$$D_{\text{amp}}(H_{g,L}) = 4J^2 \sinh(2|g|). \quad (84)$$

Therefore, for any short-time admissible normalization,

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond,\infty}(t)}{t^2} \geq \frac{2J^2 \sinh(2|g|)}{\kappa_0}. \quad (85)$$

For the tight gauge, $\kappa_0 = 1 + \varepsilon_0$.

Proof. The norm bound follows from $\|T_{\pm}\| = 1$. A bulk site has two bonds, each contributing $J^2|e^{2g} - e^{-2g}|$, giving the stated defect. Apply the preceding corollary. $\square$

VIII. LOCALITY AND BUFFERED COMPRESSED CHANNELS

At fixed time, finite-range bounded Hamiltonians generate exponentially quasi-local propagators. For the defect operators of the Julia dilation, the strict-contraction gap is what permits the functional-calculus argument to be closed in a weighted norm.

Lemma 5 (Quasi-locality of the Julia completion). *Suppose H is finite range and uniformly bounded and the normalized propagator satisfies $\|K_{H,t}\| \leq q_t < 1$ uniformly in L . Then, at each fixed finite time, $U_H(t)$, $K_{H,t}$, $D_{K_{H,t}}$, $D_{K_{H,t}^\dagger}$, and $\tilde{U}_{K_{H,t}}$ are exponentially quasi-local: there are $C_t, \mu_t > 0$ independent of L such that*

$$\|\tilde{P}_n A \tilde{P}_m\| \leq C_t e^{-\mu_t |n-m|} \quad (86)$$

for each operator in the list.

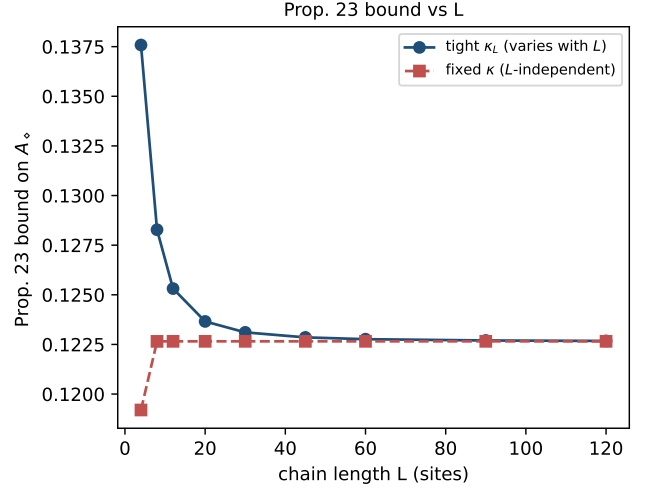


FIG. 1. Finite-size consistency check for the transition-probability lower bound. The fixed- normalization curve saturates rapidly with L at the parameter point used in the numerical study, while the tight-normalization drift is explained by the L -dependence of κ_L . The plot is a numerical consistency check, not a proof of the $t \rightarrow 0$ bound.

Proof. The finite-range propagator has a factorially decaying path expansion. To make the functional calculus explicit, introduce the weighted conjugations $A_{\sigma\mu} = e^{\sigma\mu X} A e^{-\sigma\mu X}$ for $\sigma = \pm 1$. For a finite interaction range R and uniformly bounded local matrix elements, finite range implies

$$\|H_{\sigma\mu} - H\| \leq C_H(e^{\mu R} - 1) =: \delta(\mu), \quad \delta(\mu) \rightarrow 0 \quad (\mu \rightarrow 0). \quad (87)$$

Duhamel's formula then gives a corresponding bound $\|U_{\sigma\mu}(t) - U(t)\| \leq C_t \delta(\mu)$ on each fixed finite-time interval. Because $\|K\| \leq q_t < 1$, choosing μ small enough yields weighted strict contraction $\|K_{\sigma\mu}\| \leq q_\mu < 1$ for both signs. Define $K_{\sigma\mu}^\sharp := e^{\sigma\mu X} K^\dagger e^{-\sigma\mu X}$; then $K_{\sigma\mu}^\sharp = (K_{-\sigma,\mu})^\dagger$, so it obeys the same uniform bound. Consequently, for $Z = K^\dagger K$, $Z_{\sigma\mu} = K_{\sigma\mu}^\sharp K_{\sigma\mu}$ satisfies $\|Z_{\sigma\mu}\| \leq q_\mu^2 < 1$. The absolutely convergent binomial expansion

$$(\mathbb{1} - Z)^{1/2} = \sum_{r=0}^{\infty} c_r Z^r \quad (88)$$

therefore converges after weighted conjugation, and the same argument applied to $KK^\dagger$ gives the complementary defect. This produces a uniform finite weighted norm for both defect operators and hence exponential off-diagonal decay. The constants are independent of L because the range and norm bounds entering C_H and q_t are uniform. $\square$

Proposition 11 (Dynamical transition-amplitude wit-

ness). For the canonical Julia channel,

$$A_\diamond(\mathcal{N}_{H,t}) \geq \frac{1}{2} \max_m \sum_n \left| |K_{H,t}(n,m)|^2 - |K_{H,t}(m,n)|^2 \right|. \quad (89)$$

The quantity in Eq. (89) is an operationally accessible basis-probe lower bound and is the object evaluated in the supplied numerical script. Conditional physical transition probabilities are normalization independent:

$$P_t(n|m) = \frac{|U_H(t)_{nm}|^2}{\sum_r |U_H(t)_{rm}|^2}. \quad (90)$$

For an imaginary-gauge family $H_g = S_g H_0 S_g^{-1}$ with $H_0^T = H_0$,

$$\frac{U_g(t)_{nm}}{U_g(t)_{mn}} = e^{-2g(n-m)}, \quad (91)$$

whenever the ratio is defined. These transition and reflection asymmetries are diagnostics, not additional QRT monotones.

Definition 13 (Compressed-contraction Julia channel). For a spatial window V and a strict contraction K on the full chain, define the compression

$$K_V = P_V K P_V|_{\mathcal{H}_V}, \quad \Lambda_V = \mathcal{N}_{K_V}. \quad (92)$$

Since $\|K_V\| \leq \|K\| < 1$, Λ_V is again a well-defined bistochastic Julia channel.

Proposition 12 (Buffered convergence of the compressed channel). Let H_L be finite range R , $\|H_L\| \leq h$, let $W \supset V$ contain a buffer of thickness $b > R$, and let $H_W = P_W H_L P_W$. Use the same scalar normalization for the full and buffered propagators and assume

$$\|U_L(t)\|, \|U_W(t)\| \leq q\sqrt{\kappa(t)}, \quad q < 1, \quad (93)$$

on a compact time interval with $\inf_t \kappa(t) > 0$. Then, for any fixed weighted exponent $\mu > 0$ small enough,

$$\left\| K_V^{(L)} - K_V^{(W)} \right\| \leq C_U e^{-\mu b}, \quad (94)$$

and

$$\begin{aligned} \left\| \Lambda_V^{(L)} - \Lambda_V^{(W)} \right\|_\diamond &\leq C_\Lambda e^{-\mu b}, \\ \left\| (\Lambda_V^{(L)})^{\text{rev}} - (\Lambda_V^{(W)})^{\text{rev}} \right\|_\diamond &\leq C_\Lambda e^{-\mu b}. \end{aligned} \quad (95)$$

Consequently,

$$|A_\diamond(\Lambda_V^{(L)}) - A_\diamond(\Lambda_V^{(W)})| \leq C_\Lambda e^{-\mu b}. \quad (96)$$

Proof. Write $\Delta = H_L - (H_W \oplus 0)$ and let Q project onto the boundary layer of W that is within range R of its complement. Since $b > R$, a Duhamel expansion gives

$$\|P_V(U_L(t) - U_W(t))P_V\| \leq C_U(t, \mu) e^{-\mu b}, \quad (97)$$

by combining finite-range weighted propagation with $P_V Q = 0$ and absorbing the fixed offset R into the constant. With the same scalar normalization for the two propagators, the same estimate holds for $K_V^{(L)} - K_V^{(W)}$. On the strict-contraction ball $\|K\|, \|J\| \leq q < 1$, the defect map is explicitly Lipschitz because, for $A = \mathbb{1} - K^\dagger K$ and $B = \mathbb{1} - J^\dagger J$,

$$\left\| A^{1/2} - B^{1/2} \right\| \leq \frac{\|A - B\|}{2\sqrt{1 - q^2}}, \quad (98)$$

and $\|A - B\| \leq 2q\|K - J\|$. The same bound applies to the complementary defect, so $\|\tilde{U}_K - \tilde{U}_J\| \leq C_q\|K - J\|$ for a constant depending only on q . Finally, for the associated unitary channels, $\|\mathcal{U} - \mathcal{V}\|_\diamond \leq 2\|U - V\|$, yielding the stated $e^{-\mu b}$ bounds. Transposition preserves the same weighted estimate, so the reciprocal channels obey the identical bound. $\square$

Remark 6 (Important limitation). The proposition concerns only the compressed single-particle channel Λ_V . A genuine local reduced channel

$$\mathfrak{R}_V[\mathcal{N}_L](\rho_V) = \text{Tr}_{\bar{V}}[\mathcal{N}_L(\rho_V \otimes \mathbb{1}_{\bar{V}}/d_{\bar{V}})] \quad (99)$$

requires an independently supplied tensor-product Hilbert-space realization. The native single-particle space is a direct sum, not a canonical spatial tensor product. We therefore make no claim that Λ_V equals, approximates, or bounds such a many-body reduced channel. Likewise, the covariance-matrix expression $d_V^{-1} K_{V\bar{V}} K_{V\bar{V}}^\dagger$ is a separate affine diagnostic and is not a component of a CPTP reduced channel.

IX. NUMERICAL CHECKS AND REPRODUCIBILITY STATUS

The supplied scripts evaluate only quantities that are unambiguously defined by the native single-particle construction. They are numerical consistency checks, not independent proofs of the analytic results. In particular, the Plan-A buffer script computes only the compressed-contraction channel Λ_V and the separate leakage diagnostic Ξ_V ; it does not construct a Fock-space lift, a genuine many-body density-matrix evolution, or a partial-trace reduced channel $\mathfrak{R}_V$.

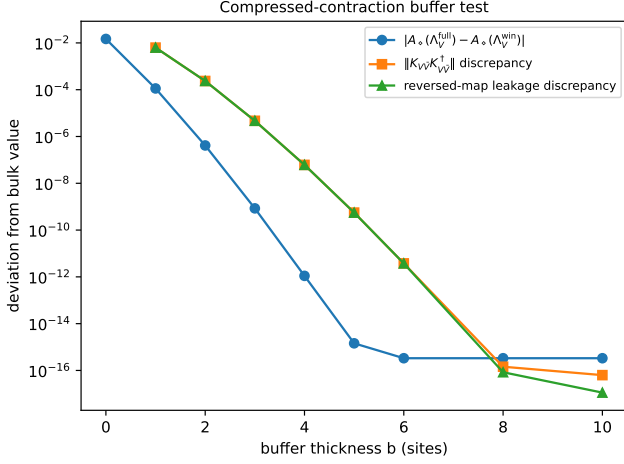


FIG. 2. Buffered convergence of the compressed-contraction Julia channel $\Lambda_V = \mathcal{N}_{P_V K P_V}$ for the numerical parameter point $(g, t, \ell, L) = (0.5, 0.35, 4, 60)$. The asymmetry difference reaches the floating-point floor rapidly. The leakage curves show the corresponding un-normalized covariance-matrix coupling diagnostic; the normalized offset is not plotted because its complement-dimension prefactor introduces an unrelated $1/b$ effect. No numerical decay exponent is claimed.

A. Quasi-locality check

At $L = 60$, $g = 0.5$, and $t = 0.35$, the script gives the effective slopes

$$\begin{aligned} \mu(U_H) &= \mu(K) \approx 2.5648, \\ \mu(D_K) &\approx \mu(D_{K^\dagger}) \approx 0.6625. \end{aligned} \quad (100)$$

over the short fitting window used in the code. Re-fitting the defect operators over a longer window gives $\mu \approx 0.5938$, demonstrating the expected fit-window dependence. These are finite-size effective slopes, not claimed universal locality exponents.

B. Single-particle thermodynamic persistence check

For $g = 0.5$, $J = 1$, $m = 0$, and $t = 0.35$, with $L_{\text{bulk}} = 60$, $\ell = 4$, and central window V chosen as in the supplied script, the common normalization is $\kappa = 2.286505$. The Proposition 89 lower bound with this L -independent normalization approaches

$$0.119198, 0.122654, 0.122656, \dots \quad (101)$$

for $L = 4, 8, 12, \dots$, and is numerically constant to six digits beyond $L \approx 12$. With the per-size tight normalization the values drift from 0.137583 at $L = 4$ to 0.122676 at $L = 120$, while the product κ_L times the bound is approximately constant, 0.280454, for $L \geq 8$. This identifies the visible drift as a normalization effect rather than evidence against single-particle thermodynamic persistence. The calculation is a finite- t consistency check and does not test the $t \rightarrow 0$ coefficient directly.

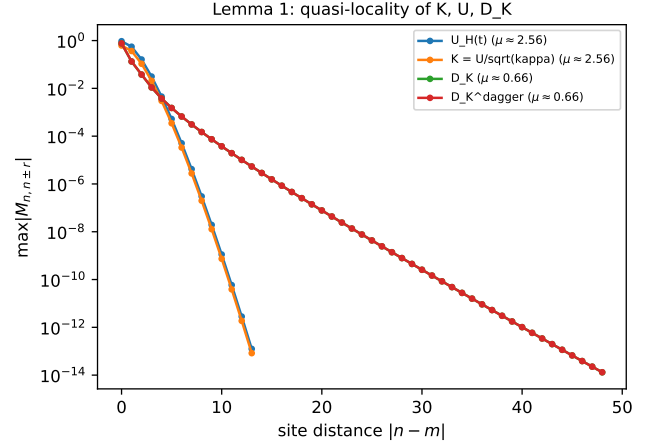


FIG. 3. Numerical consistency check of quasi-locality at $L = 60$. The propagator and scalar-normalized contraction have the same profile up to normalization, while the Julia defect operators decay more slowly and are more sensitive to the fit window. The analytic statement is the existence of an L -uniform exponential bound; the figure does not prove that statement.

C. Buffered compressed-channel check

For the same parameter point and $\ell = 4$, the full-chain compressed-channel asymmetry is

$$A_\diamond(\Lambda_V^{(L)}) = 0.528469. \quad (102)$$

The isolated-buffer values differ by

$$1.13 \times 10^{-4}, \quad 4.14 \times 10^{-7}, \quad 8.47 \times 10^{-10}, \quad 1.10 \times 10^{-12} \quad (103)$$

for $b = 1, 2, 3, 4$, respectively, before reaching double-precision noise. The un-normalized leakage difference is approximately

$$6.30 \times 10^{-3}, \quad 2.36 \times 10^{-4}, \quad 4.69 \times 10^{-6}, \quad 6.12 \times 10^{-8}, \dots, \quad (104)$$

while the normalized covariance diagnostic displays an approximately $1/b$ trend because the full-chain and window calculations use different complement dimensions. Accordingly, the numerical study does not claim exponential convergence of the normalized offset.

Remark 7 (What the scripts do not test). *Neither script computes the genuine many-body reduced channel discussed above. The first computes Λ_V and a separate leakage diagnostic; the second tests quasi-locality and the transition-amplitude lower bound. This distinction is part of the mathematical scope of the paper.*

X. RELATION TO STANDARD NHSE DIAGNOSTICS AND LIMITATIONS

The channel QRT is primary. Spectral winding, GBZ radius, transfer-matrix Lyapunov exponents, edge-

concentration measures, and dynamical reflection asymmetries are diagnostics of particular Hamiltonian families rather than additional free-operation monotones.

For reciprocal disordered chains with invertible nearest-neighbor blocks, the transfer cocycle can be symplectically gauged so that singular values occur in reciprocal pairs, yielding

$$\gamma_j(E) = -\gamma_{2M+1-j}(E). \quad (105)$$

For the clean constant-transfer case, $\gamma_j = (1/a) \ln |\beta_j|$, so the usual GBZ crossing is $\gamma_M = \gamma_{M+1}$. These results provide consistency relations between reciprocity and standard non-Bloch/transfer diagnostics, but they do not promote those quantities to resource monotones.

A useful sharpness example is the direct sum

$$H_{\oplus,g}(k) = H_g(k) \oplus H_{-g}(k), \quad g \neq 0. \quad (106)$$

The two sectors have opposite skin accumulation and opposite determinant windings, so $W_{\oplus,g} = 0$, while $H_{\oplus,g}^T \neq H_{\oplus,g}$ and hence $A_{\diamond}^{\text{sup}}(H_{\oplus,g}) > 0$. In a species-twisted reciprocity frame the same direct sum is reciprocal, illustrating explicitly that the resource is a frame-dependent time-reversal asymmetry. This is why a broader symmetry architecture would require an explicitly supplied involution r , rather than an unqualified claim of universality.

The particular global monotones used here are bounded by one, so they are persistence-type quantities rather than extensive tensor-power rates. This statement concerns these max-type quantities, not asymmetry measures in general. An extensive quantity can still be defined operationally for an explicit tensor-product array of independent cells by summing per-cell asymmetries, but that is a different architecture from a globally normalized single-particle direct sum and is not needed for the main results.

Finally, a genuinely open-system Lindbladian resource theory would require an enlarged primitive object class. The present Julia-completed channel should not be identified with arbitrary gain/loss Lindblad dynamics or with every microscopic realization of NHSE [20].

XI. CONCLUSION

We have constructed a fixed-time channel resource theory of transpose-reciprocity breaking for normalized non-Hermitian propagators represented by a canonical Julia completion. The central QRT objects are CPTP bistochastic channels, free channels are transpose reciprocal, and free superchannels are physically realizable, object-preserving, transpose-covariant superchannels. The principal monotones are

$$A_{\diamond}(\mathcal{N}) = \frac{1}{2} \|\mathcal{N} - \mathcal{N}^{\text{rev}}\|_{\diamond} \quad (107)$$

and generalized robustness. The first has the operational discrimination meaning $A_{\diamond} = 2p_{\text{succ}}^{\text{opt}} - 1$.

The canonical Julia representative obeys

$$D_K^T = D_{K^*}, \quad D_{K^\dagger}^T = D_{K^T}, \quad \tilde{U}_K^T = \tilde{U}_{K^T}, \quad (108)$$

so its reciprocity support is exactly the condition $K^T = K$. The quantitative resource value is normalization dependent, but the zero-versus-positive support is not within the admissible strict scalar class. Under the per-Hamiltonian tight gauge, the trajectory-derived supremum is invariant under overall positive energy rescaling.

The two-site Hatano–Nelson orbit provides the exact calibration

$$A_{\diamond}(\mathcal{N}_{g,\tau_*}) = \sqrt{1 - c(g)^2}, \quad (109)$$

with strict monotonicity in $|g|$. Hence every deterministic free conversion obeys the necessary condition $|g| \geq |g'|$, while the explicitly restricted unitary-congruence subclass is rigid. Whether the full ancilla-assisted free-superchannel class admits deterministic attenuation remains open.

For uniformly bounded finite-range chains, an L -uniform short-time estimate gives

$$A_{\diamond}(\mathcal{N}_{H_L,t}) \geq \frac{t^2}{2\kappa_0} D_{\text{amp}}(H_L) - O(t^3), \quad (110)$$

with an L -independent remainder. For the uniform Hatano–Nelson family,

$$\liminf_{t \rightarrow 0^+} \frac{A_{\diamond,\infty}(t)}{t^2} \geq \frac{2J^2 \sinh(2|g|)}{1 + \varepsilon_0} > 0 \quad (g \neq 0). \quad (111)$$

On the same uniform Hatano–Nelson family,

$$\boxed{A_{\diamond}^{\text{sup}}(H_g) > 0 \iff g \neq 0 \iff \rho_{\text{GBZ}} > 0 \iff \text{conventional OBC NHSE.}} \quad (112)$$

which is the principal exact NHSE witness result of the paper.

For coupled chains, the rigorous local result is instead

$$\Lambda_V = \mathcal{N}_{P_V K P_V}, \quad (113)$$

whose comparison to an isolated buffered window is exponentially controlled. No many-body reduced-channel theorem is claimed, because the native single-particle direct sum carries no canonical spatial tensor-product factorization. The numerical scripts are consistent with these analytic results and, importantly, do not replace the compressed channel by a fictitious partial trace construction.

The appropriate interpretation is therefore precise rather than universal: the paper supplies an axiomatic symmetry QRT for transpose-reciprocity breaking, an exact Hatano–Nelson calibration sector, and a uniform short-time persistence theorem in the single-particle thermodynamic limit. Universal coverage of all NHSE mechanisms, a microscopic hardware theory of free operations, deterministic full-class attenuation, and the relation to genuine many-body reduced channels remain open problems.

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